

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 5 - ORDINARY DIPLOMA IN ENGINEERING

CONTINUOUS ASSESSMENT TEST - I - NOVEMBER-2024

EDT 05103 - APPLIED CHEMISTRY

SEMESTER: I

DURATION: 1 Hr 30 Min.

YEAR: II

MAX. MARKS: 50

PART - A (Answer all the Questions) (7 x 2 = 14)

1. What is material science and material engineering?
2. Why do we study material science?
3. Mention three criteria that are important in the materials selection process
4. What is Faraday?
5. What is the electrochemical equivalent?
6. Define the term electrode potential
7. Differentiate batteries from cells

PART - B (Answer any three Questions) (3 x 4 = 12)

8. How do grain boundaries influence the mechanical properties of materials?
9. Mention the levels through which the structural elements may be classified on the basis of size.
10. What is the difference between single crystal, polycrystalline and amorphous solids?
11. Briefly explain the concept of nanotechnology as it applies to materials
12. It is not possible to create such a perfect material that is free from lattice irregularity, what are the crystalline defects that may arise in materials?
13. List down at least three characteristics of electrochemical series

PART - C (Answer any two Questions) (2 x 12 = 24)

14. Write down at least eight (8) applications of electrochemical series.
15. Daniel cell is a typical example of Galvanic cell. Draw and describe the EMF of a Daniel cell
16. Some metals and chemicals are produced by electrochemical methods, with the aid of a diagram briefly explain the Electrolysis process of a molten sodium chloride (NaCl)
17. i. What are Plastics?
 - ii. Explain types of polymerization process as applied in manufacturing of plastics
 - iii. With examples, give the difference(s) between Thermoplastic and Thermosetting plastics.

All The Best